PROGRAM:

```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0

    while base < total_frames:
        window_end = min(base + window_size - 1,
                         total_frames - 1)

        print("\n" + "-" * 50)
        print(f"Sender Window: [Frame {base} to
              Frame {window_end}]")

        # send all frames within the current window that
        haven't been sent
        while next_seq < base + window_size and
              next_seq < total_frames:
            print(f"Node A --> Sending Frame {next_seq}")
            if next_seq == lost_frame:
                print(f"(Frame {next_seq} LOST in
                      transit!)")
            next_seq += 1

        if lost_frame is not None and base <= lost_frame
                                    < next_seq:
            print(f"Node B --> Frame {lost_frame} is
                  missing, Discarding out-of-order frames
                  after it")
            print(f"Node A --> Timeout! Resending
                  window starting from Frame {lost_frame}")
            next_seq = lost_frame
            lost_frame = None
        continue
```

Ex 7 Flow Control at Data Link Layer using Sliding Window Protocol.

AIM:

To write a program to implement flow control at Data Link Layer using the Sliding Window Protocol, and to simulate the flow of frames from one node to another.

ALGORITHM:

1. Start the program
2. Read the total no. of frames to be sent (N), the sender's window size (W), (optionally) a frame no to simulate as lost during transmission.
3. Initialize base = 0 (sequence no. of oldest unacknowledged frame) and nextseq = 0
4. Repeat while base < N:

a) Determine the current sender window as the range [base, base + W - 1]

b) Send all frames with sequence numbers from nextseq up to (base + W - 1), incrementing nextseq after each from Since the window restricts the sender to at most W outstanding (unacknowledged) frames at any time.

c) If the lost frame lies within the frame just sent:

(i) The receiver accepts each frame in order

(ii) On receiving ACK for the last frame sent.

5) Repeat step 4 until all N frames have been transmitted & acknowledged

6) Stop the program

```
    for f in range(base, next_seq):
        print(f"Node B --> Received frame {f},
                Sending ACK {f}")
    print(f"Node A --> ACK {next_seq - 1} received.
    Sliding window forward.")
    base = next_seq
    print("\n" + "-" * 50)
print(f"All {total_frames} frames
    successfully transmitted from Node A to Node
def main():
    print("=== SLIDING WINDOW PROTOCOL - FLOW
            CONTROL SIMULATION (Go-Back-N) ===\n")
    total_frames = int(input("Enter total no. of
                frames to send: "))
    window_size = int(input("Enter sender's window
    lost_frame = int(input("Enter frame number to
            simulate as LOST\n(0 to {total_frames-1},
            enter -1 for no loss): "))
    if lost_frame == -1:
        lost_frame = None
    simulate(total_frames, window_size, lost_frame)
if __name__ == "__main__":
    main()
```

OUTPUT:

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION ===

Enter total number of frames to send: 8
Enter sender's window size: 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss): 5

--------------------------------------------------

Sender window: [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received, sliding window forward

--------------------------------------------------

Sender window: [Frame 4 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward

--------------------------------------------------

All 8 frames successfully transmitted from Node A to B
```

Result: The program to implement flow control at Data Link Layer using Sliding Window Protocol was successfully executed